

Mashuk Elahi

PERSONAL INFORMATION

Name	Mashuk Elahi
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DESIRED EMPLOYMENT / OCCUPATIONAL FIELD

Electrical Engineer — Mission-Critical Power Systems & Critical Facilities

16 years commissioning mission-critical power systems — MV 11kV switchgear, UPS, BESS, IEC 61850/SCADA protection — in 24/7 zero-downtime environments. Delivers the full commissioning lifecycle from Factory Testing (Level 1) through Closeout/Handover (Level 6) across MV/LV power distribution, EPMS/SCADA monitoring, and BESS integration. Certified ISO 9001:2015 & ISO 50001:2018 Lead Auditor (CQI-IRCA) with Python automation experience scaling EPMS load-list and cable-schedule verification.

WORK EXPERIENCE

07/2024 - Present

Electrical Engineer — Mission-Critical Power Systems

Strategic Marine Pte. Ltd. • Singapore

- Architected the complete LV power distribution system for a multi-generator heavy-industry power plant (3 prime-power generator sets, 450 ekW total) — single-line diagrams, load lists, and cable schedules — securing first-pass approval across four independent certification bodies with zero rework cycles.
- Integrated a 72kWh Li-ion Battery Energy Storage System (BESS), BMS, and DC/DC inverters into a hybrid power architecture, cutting grid/fuel energy draw by up to 50% during low-load operation; led BMS-to-EPMS interface integration and Factory Acceptance Testing (FAT) with third-party certification.
- Delivered full electrical design across a multi-site heavy-industry power programme — dual 380/220V AC + 24V DC distribution, interlocked standby power systems, approved across four independent certification bodies.

Sector: Heavy-Industry Power Systems

02/2023 - 07/2024

Senior Project Engineer — Semiconductor & High-Tech Facilities

Exyte • Singapore

- Led 300mm semiconductor cleanroom critical-power hook-up and energisation to IEC 60204/61511/ISO 14644 — zero audit rework against 100% tool-uptime / N+1 availability targets; PLC/SCADA interlock and functional performance testing cleared first pass.
- Introduced prefabricated power skids that cut on-site installation labour by ~50% and eliminated contamination risk during live energised operations.
- Drove ISO 50001 energy-efficiency stewardship through smart EPMS load monitoring as certified Lead Auditor.

Sector: Semiconductor / Critical Facilities

01/2021 - 01/2023

E&I Engineer — Hazardous-Area, Power & Controls

Phoenix Mecano (S.E.A.) Pte Ltd • Singapore

- Pioneered Malaysia's first solar-wind autonomous remote power installation (24V DC solar/battery bus) — eliminated routine diesel generator use and enabled full remote/unmanned operation.

- Directed an onshore cogeneration plant electrical & instrumentation scope — transformers, MV/LV switchgear, 400kVA UPS; SCADA integration and technical bid evaluation for all major electrical packages.
- Designed IECEx/ATEX Zone 1/2 hazardous-area systems and standardised compliance documentation as certified Lead Auditor.

Sector: Industrial Power Systems

11/2012 - 12/2020

E&I Superintendent — Heavy-Industry Power Systems

Keppel FELS · Singapore

- Superintended crews of up to 100+ across fabrication, installation, and integrated commissioning of large-scale heavy-industry power systems — energised MV 11kV switchgear and ABB 800xA SCADA-based fire & gas protection; maintained MV/LV power reliability at 100% uptime through final systems turnover.
- Energised a wind-power substation electrical system to 100% IEC 61850 protection-relay conformity; zero punch-list items outstanding at handover.
- Commissioned large-scale mobile heavy-industry power and accommodation modules through FMEA and acceptance testing; mentored 15+ engineers via Innovation & Quality Circles.

Sector: Heavy-Industry Power Systems

09/2009 - 10/2012

Assistant Electrical Engineer — Industrial & Infrastructure Projects

Ananda Shipyard & Slipways Ltd · Bangladesh

- Led electrical & instrumentation crews of 15-30 on newbuild heavy-industry power projects, achieving first-time milestone sign-off.
- Delivered LV/MV power distribution, protection coordination, and commissioning across industrial and infrastructure projects — establishing the foundation of 16 years' uninterrupted career in electrical engineering.

Sector: Industrial & Infrastructure

EDUCATION AND TRAINING

2026

MSc Sustainable Electricity Generation

Universidad Católica de Murcia (UCAM), Spain

2026

MSc Digital Transformation

London Metropolitan University, UK

2025

Master of Business Administration (MBA)

University of Sunderland, UK

2008

BSc Electrical & Electronic Engineering

Bangladesh University

2004

Diploma in Electronics Engineering

Jessore Polytechnic Institute

LANGUAGE SKILLS

Self-assessment, Common European Framework of Reference for Languages (CEFR)

Language	Listening	Reading	Spoken production	Spoken interaction	Writing
Bengali — Mother tongue	—	—	—	—	—
English	C2	C2	C2	C2	C2

DIGITAL SKILLS

- Electrical design & modelling: AutoCAD Electrical, Navisworks 3D, ETAP, SKM PowerTools, MathCAD, MATLAB/Simulink
- Controls & SCADA platforms: ABB 800xA, Siemens & Yokogawa PLC, IEC 61850 protection systems

- Engineering automation: Python-based EPMS load-list & cable-schedule automation, AI agent workflow integration
- General office & collaboration: Microsoft Office, SAP (BOM), Fluke DSX2-8000 cable test/reporting software

OTHER SKILLS

- Cross-discipline team leadership (crews of 100+); mentoring of 15+ engineers via Innovation & Quality Circles
- Cross-cultural team collaboration across Singapore, Bangladesh, and multinational project environments
- Certified Lead Auditor — ISO 9001:2015 & ISO 50001:2018 (CQI-IRCA)
- Risk & reliability methods: FMEA, HAZID, cause-and-effect analysis, Tier III/IV design intent, N+1 / 2N+1 redundancy

CERTIFICATIONS & TRAINING

- CDCP (Certified Data Centre Professional) — in progress
- ISO 50001:2018 Energy Management — Lead Auditor (CQI-IRCA)
- ISO 9001:2015 Quality Management — Lead Auditor (CQI-IRCA)
- IECEx Ex 001 — Basic Principles of Protection in Explosive Atmospheres (valid to Oct 2027)
- ABB — Power System / LV Switchboard Training
- Fluke DSX2-8000 Cable Analyzer Copper Certifier Training